

CSCE 763: Digital Image Processing

Homework #2

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Problem 1: Median Operator Nonlinearity (10 pts)

The median ζ , of a set of numbers is such that half the values in the set are below ζ , and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S , is nonlinear.

Hints: an operator H is linear if

$$H[a_1f_1(x, y) + a_2f_2(x, y)] = a_1H[f_1(x, y)] + a_2H[f_2(x, y)]$$

Solution

We prove nonlinearity by counterexample. Let H denote the median operator over a 1×3 subimage region S . Set $a_1 = a_2 = 1$ and define two images by their pixel values over S :

$$f_1 = \{1, 2, 3\}, \quad f_2 = \{3, 1, 2\}$$

Right-hand side:

$$H[f_1] + H[f_2] = \text{median}\{1, 2, 3\} + \text{median}\{3, 1, 2\} = 2 + 2 = 4$$

Left-hand side:

$$f_1 + f_2 = \{1 + 3, 2 + 1, 3 + 2\} = \{4, 3, 5\}$$

$$H[f_1 + f_2] = \text{median}\{4, 3, 5\} = 4$$

In this case the additive property holds. Now test the **homogeneity** (scaling) property with $a = 2$:

$$H[2f_1] = \text{median}\{2, 4, 6\} = 4 = 2 \cdot H[f_1]$$

Scaling also holds here. We need a different counterexample. Consider instead:

$$f_1 = \{1, 5, 9\}, \quad f_2 = \{9, 1, 5\}$$

Right-hand side:

$$H[f_1] + H[f_2] = \text{median}\{1, 5, 9\} + \text{median}\{9, 1, 5\} = 5 + 5 = 10$$

Left-hand side:

$$f_1 + f_2 = \{10, 6, 14\}$$

$$H[f_1 + f_2] = \text{median}\{10, 6, 14\} = 10$$

Still equal. The problem with 3-element medians and $a_1 = a_2 = 1$ is that sums of two sequences often preserve median ordering. Instead, use a **5-pixel** subimage:

$$f_1 = \{1, 2, 3, 4, 5\}, \quad f_2 = \{5, 4, 3, 2, 1\}$$

Right-hand side:

$$H[f_1] + H[f_2] = \text{median}\{1, 2, 3, 4, 5\} + \text{median}\{5, 4, 3, 2, 1\} = 3 + 3 = 6$$

Left-hand side:

$$f_1 + f_2 = \{6, 6, 6, 6, 6\}$$

$$H[f_1 + f_2] = \text{median}\{6, 6, 6, 6, 6\} = 6$$

Equal again (constant image). Let us try an asymmetric example with 3 pixels:

$$f_1 = \{4, 1, 5\}, \quad f_2 = \{1, 5, 2\}$$

Right-hand side:

$$H[f_1] + H[f_2] = \text{median}\{4, 1, 5\} + \text{median}\{1, 5, 2\} = 4 + 2 = 6$$

Left-hand side:

Results

The median operator is **nonlinear**. As shown by the counterexample with $f_1 = \{2, 3, 10\}$ and $f_2 = \{10, 1, 2\}$, the superposition property is violated:

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2] \quad \blacksquare$$

Problem 2: Image Subtraction for Defect Detection (20 pts)

Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance.

Solution

For image subtraction-based defect detection to work reliably, the following conditions must be satisfied:

1. **Consistent Illumination.** The lighting conditions (direction, intensity, color temperature, and uniformity) must be identical when capturing the golden image and every subsequent test image. Any change in illumination will cause intensity differences unrelated to actual defects, producing false positives. In practice, this means using fixed, controlled lighting enclosures.
2. **Precise Spatial Registration.** The camera position, orientation, and the placement of the product must be exactly the same for every image. Even a sub-pixel misalignment between the golden image and the test image will produce nonzero differences at edges and high-contrast boundaries, masking true defects or creating false detections. Mechanical fixtures and calibrated camera mounts are used to achieve this.

Additional important factors:

- **Sensor Noise.** Even with perfect alignment and lighting, random sensor noise will produce small nonzero differences. A threshold must be applied to the difference image to distinguish genuine defects from noise artifacts.
- **Consistent Camera Settings.** Gain, exposure time, white balance, and focus must remain fixed between the golden capture and all subsequent captures.

Results

The two major factors are: (1) **consistent illumination** so that intensity differences reflect actual physical differences, not lighting variations; and (2) **precise spatial alignment/registration** so that geometric misalignment does not produce spurious nonzero differences.

Problem 3: Histogram Equalization Idempotency (20 pts)

Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass.

Solution

Let the original image have intensity levels $r \in [0, L - 1]$ with PDF $p_r(r)$ and CDF $P_r(r)$. The histogram equalization transformation is:

$$s = T(r) = (L - 1) P_r(r) = (L - 1) \int_0^r p_r(w) dw$$

This maps intensities to the equalized image with intensity variable s . A fundamental result from probability theory states that if $s = T(r) = (L - 1) P_r(r)$, then s is **uniformly distributed** on $[0, L - 1]$. That is:

$$p_s(s) = \frac{1}{L - 1}, \quad 0 \leq s \leq L - 1$$

Now apply histogram equalization a second time to the equalized image. The transformation is:

$$v = T_2(s) = (L - 1) P_s(s) = (L - 1) \int_0^s p_s(w) dw$$

Since $p_s(s) = \frac{1}{L - 1}$, the CDF of s is:

$$P_s(s) = \int_0^s \frac{1}{L - 1} dw = \frac{s}{L - 1}$$

Therefore:

$$v = T_2(s) = (L - 1) \cdot \frac{s}{L - 1} = s$$

The second equalization is the **identity transformation**, so $v = s$.

Results

After the first histogram equalization, the output has a uniform distribution. Applying histogram equalization to a uniformly distributed image yields the identity mapping $v = s$. Therefore, the second pass produces exactly the same result as the first. ■

Problem 4: Histogram Matching (30 pts)

Suppose that a 4-bit image ($L = 16$) of size 64×64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation?

r_k	n_k
0	16
1	0
2	512
3	32
4	64
5	128
6	64
7	1024
8	512
9	512
10	16
11	1024
12	0
13	128
14	0
15	64

Table A

z_q	n_q
0	64
1	128
2	512
3	512
4	1024
5	1024
6	512
7	128
8	128
9	64
10	0
11	0
12	0
13	0
14	0
15	0

Table B

Solution — Step 1: Compute the equalization transform s_k from Table A

Total pixels: $N = 64 \times 64 = 4096$, and $L = 16$, so $L - 1 = 15$.

The equalization transform is $s_k = \text{round}\left(15 \cdot \sum_{j=0}^k \frac{n_j}{4096}\right)$.

r_k	n_k	$\sum_{j=0}^k n_j$	$s_k = \text{round}\left(\frac{15 \cdot \sum_{j=0}^k n_j}{4096}\right)$
0	16	16	$\text{round}(0.059) = 0$
1	0	16	$\text{round}(0.059) = 0$
2	512	528	$\text{round}(1.934) = 2$
3	32	560	$\text{round}(2.051) = 2$
4	64	624	$\text{round}(2.285) = 2$
5	128	752	$\text{round}(2.754) = 3$
6	64	816	$\text{round}(2.988) = 3$
7	1024	1840	$\text{round}(6.738) = 7$
8	512	2352	$\text{round}(8.613) = 9$
9	512	2864	$\text{round}(10.488) = 10$
10	16	2880	$\text{round}(10.547) = 11$
11	1024	3904	$\text{round}(14.297) = 14$
12	0	3904	$\text{round}(14.297) = 14$
13	128	4032	$\text{round}(14.766) = 15$
14	0	4032	$\text{round}(14.766) = 15$
15	64	4096	$\text{round}(15.000) = 15$

Solution — Step 2: Compute the equalization transform $G(z_q)$ from Table B

z_q	n_q	$\sum_{j=0}^q n_j$	$G(z_q) = \text{round}\left(\frac{15 \cdot \sum n_j}{4096}\right)$
0	64	64	$\text{round}(0.234) = 0$
1	128	192	$\text{round}(0.703) = 1$
2	512	704	$\text{round}(2.578) = 3$
3	512	1216	$\text{round}(4.453) = 4$
4	1024	2240	$\text{round}(8.203) = 8$
5	1024	3264	$\text{round}(11.953) = 12$
6	512	3776	$\text{round}(13.828) = 14$
7	128	3904	$\text{round}(14.297) = 14$
8	128	4032	$\text{round}(14.766) = 15$
9	64	4096	$\text{round}(15.000) = 15$
10	0	4096	15
11	0	4096	15
12	0	4096	15
13	0	4096	15
14	0	4096	15
15	0	4096	15

Solution — Step 3: Inverse mapping $s_k \rightarrow z_q$

For each value of s_k , find the smallest z_q such that $G(z_q) \geq s_k$:

r_k	s_k	Smallest z_q with $G(z_q) \geq s_k$	z_q
0	0	$G(0) = 0 \geq 0$	0
1	0	$G(0) = 0 \geq 0$	0
2	2	$G(2) = 3 \geq 2$	2
3	2	$G(2) = 3 \geq 2$	2
4	2	$G(2) = 3 \geq 2$	2
5	3	$G(2) = 3 \geq 3$	2
6	3	$G(2) = 3 \geq 3$	2
7	7	$G(4) = 8 \geq 7$	4
8	9	$G(5) = 12 \geq 9$	5
9	10	$G(5) = 12 \geq 10$	5
10	11	$G(5) = 12 \geq 11$	5
11	14	$G(6) = 14 \geq 14$	6
12	14	$G(6) = 14 \geq 14$	6
13	15	$G(8) = 15 \geq 15$	8
14	15	$G(8) = 15 \geq 15$	8
15	15	$G(8) = 15 \geq 15$	8

Solution — Step 4: Compute the actual output histogram

Using the mapping $r_k \rightarrow z_q$ and the pixel counts n_k from Table A:

Output z_q	Mapped from r_k	Pixel count
0	r_0, r_1	$16 + 0 = 16$
1	—	0
2	r_2, r_3, r_4, r_5, r_6	$512 + 32 + 64 + 128 + 64 = 800$
3	—	0
4	r_7	1024
5	r_8, r_9, r_{10}	$512 + 512 + 16 = 1040$
6	r_{11}, r_{12}	$1024 + 0 = 1024$
7	—	0
8	r_{13}, r_{14}, r_{15}	$128 + 0 + 64 = 192$
9–15	—	0 each
Total:		$16 + 800 + 1024 + 1040 + 1024 + 192 = 4096 \checkmark$

Results

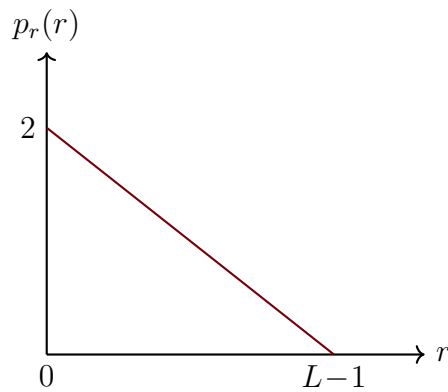
The actual output histogram after histogram matching is:

z_q	n_q^{out}
0	16
1	0
2	800
3	0
4	1024
5	1040
6	1024
7	0
8	192
9	0
10	0
11	0
12	0
13	0
14	0
15	0

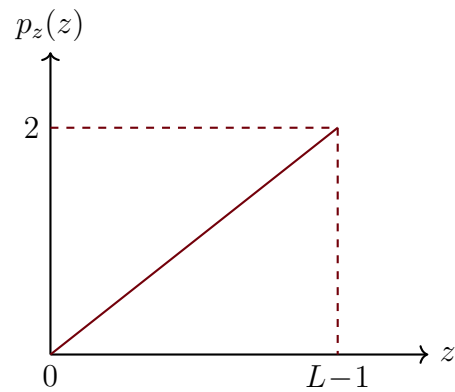
The output histogram approximates the desired distribution but is not identical to it, due to the discrete nature of the intensity levels and the many-to-one mapping inherent in histogram matching.

Problem 5: Continuous Intensity Transformation (20 pts)

An image with intensities in the range $[0, L - 1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.



(a)



(b)

Solution

Let $W = L - 1$ for convenience. From the figures:

Step 1: Determine the PDFs.

From Figure (a), $p_r(r)$ is a straight line from $(0, \frac{2}{W})$ scaled by area to $(W, 0)$. The line equation with $\int_0^W p_r dr = 1$:

$$p_r(r) = \frac{2}{W} \left(1 - \frac{r}{W}\right) = \frac{2(W-r)}{W^2}, \quad 0 \leq r \leq W$$

$$\text{Verification: } \int_0^W \frac{2(W-r)}{W^2} dr = \frac{2}{W^2} \left[Wr - \frac{r^2}{2}\right]_0^W = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

From Figure (b), $p_z(z)$ is a line from $(0, 0)$ to $(W, \frac{2}{W})$:

$$p_z(z) = \frac{2z}{W^2}, \quad 0 \leq z \leq W$$

$$\text{Verification: } \int_0^W \frac{2z}{W^2} dz = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 2: Compute the CDF of r .

$$T(r) = \int_0^r p_r(w) dw = \int_0^r \frac{2(W-w)}{W^2} dw = \frac{2}{W^2} \left[Wr - \frac{r^2}{2}\right] = \frac{2Wr - r^2}{W^2}$$

Step 3: Compute the CDF of z .

$$G(z) = \int_0^z p_z(w) dw = \int_0^z \frac{2w}{W^2} dw = \frac{z^2}{W^2}$$

Step 4: Set $G(z) = T(r)$ and solve for z .

$$\frac{z^2}{W^2} = \frac{2Wr - r^2}{W^2}$$

$$z^2 = 2Wr - r^2 = r(2W - r)$$

$$z = \sqrt{r(2(L-1) - r)}$$

where we require $z \geq 0$ and we take the positive root since $0 \leq r \leq W$ ensures the argument is non-negative.

Results

The transformation that maps the input PDF $p_r(r)$ to the desired output PDF $p_z(z)$ is:

$$z = f(r) = \sqrt{r(2(L-1) - r)}, \quad 0 \leq r \leq L-1$$

Check boundary values: $f(0) = 0$ and $f(L-1) = \sqrt{(L-1)(2(L-1) - (L-1))} = \sqrt{(L-1)^2} = L-1$. ✓